



Solution Set

Question 1

A collection is a set only when membership is well-defined. Thus the well-defined examples are months beginning with J, boys in a class, natural numbers less than 100, Prem Chand novels, even integers, and questions in the chapter. Subjective collections such as most talented writers, best batsmen, or most dangerous animals are not sets.

Question 2

Roster forms:

$$\begin{aligned} A &= \{-3, -2, -1, 0, 1, 2, 3, 4, 5, 6\} \\ B &= \{1, 2, 3, 4, 5\}, \quad C = \{17, 26, 35, 44, 53, 62, 71, 80\} \\ D &= \{2, 3, 5\}, \quad E = \{T, R, I, G, O, N, M, E, Y\}, \quad F = \{B, E, T, R\} \end{aligned}$$

Question 3

Set-builder examples:

$$\begin{aligned} \{3, 6, 9, 12\} &= \{3n : n \in \mathbb{N}, 1 \leq n \leq 4\} \\ \{2, 4, 8, 16, 32\} &= \{2^n : 1 \leq n \leq 5\} \\ \{5, 25, 125, 625\} &= \{5^n : 1 \leq n \leq 4\} \end{aligned}$$

Question 4

The empty sets are the sets of months with 32 days, whole numbers less than 0, and even primes greater than 2. The set of even prime numbers is not empty because it contains 2.

Question 5

For the subset statements: false, true, false, first part true and second part false, true.

Question 6

For

$$A = \{1, 2, \{3, 4\}, 5\}$$

the truth values are: false, true, true, true, false, true, false, false, false, true, false.

Question 7

Power sets:

$$\begin{aligned}\mathcal{P}(\{a\}) &= \{\emptyset, \{a\}\} \\ \mathcal{P}(\{a, b\}) &= \{\emptyset, \{a\}, \{b\}, \{a, b\}\} \\ \mathcal{P}(\emptyset) &= \{\emptyset\}\end{aligned}$$

Question 8

The intervals are

$$(-4, 6], \quad (-12, -10), \quad [0, 7), \quad [3, 4]$$

Question 9

Set-builder forms are

$$\{x \in \mathbb{R} : -3 < x < 0\}, \quad \{x \in \mathbb{R} : 6 \leq x < 12\}, \quad \{x \in \mathbb{R} : -23 \leq x \leq 5\}$$

Question 10

Using the definitions:

$$\begin{aligned}A \cap B &= B, \quad A \cap C = C, \quad A \cap D = D \\ B \cap C &= \emptyset, \quad B \cap D = \{2\}, \quad C \cap D = \{\text{odd primes}\}\end{aligned}$$

Question 11

Complements in the universal set give

$$\begin{aligned}A' &= \{1, 3, 5, 7, 9\}, \quad B' = \{1, 4, 6, 8, 9\} \\ (A \cup B)' &= A' \cap B' = \{1, 9\} \\ (A \cap B)' &= A' \cup B' = \{1, 3, 4, 5, 6, 7, 8, 9\}\end{aligned}$$

Question 12

The quadratic factorizes as

$$x^2 - 8x + 12 = (x - 2)(x - 6)$$

so

$$A = \{2, 6\}$$

and the containment chain is

$$D \subset A \subset B \subset C$$

Question 13

The false statements can be disproved by counterexample; the true one is transitivity:

$$A \subset B, \quad B \subset C \Rightarrow A \subset C$$

Question 14

If

$$A \cap B = A \cap C, \quad A \cup B = A \cup C$$

then every element of B is in C and every element of C is in B. Therefore

$$B = C$$

Question 15

Equivalent subset conditions:

$$A \subset B \iff A - B = \emptyset \iff A \cup B = B \iff A \cap B = A$$

Question 16

If

$$A \subset B$$

then any element of

$$C - B$$

is not in B and hence not in A. Therefore

$$C - B \subset C - A$$

Question 17

Using disjoint decomposition:

$$A = (A \cap B) \cup (A - B)$$

$$A \cup (B - A) = A \cup B$$

Question 18

The absorption identities are

$$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$$

Question 19

A counterexample is

$$A = \emptyset, \quad B = \{1\}, \quad C = \{2\}$$

Then

$$A \cap B = A \cap C = \emptyset$$

but

$$B \neq C$$

Question 20

If

$$A \cup X = B \cup X, \quad A \cap X = B \cap X = \emptyset$$

then an element of A cannot be in X, so it must be in B; symmetry gives

$$A = B$$